**R13**

**Code No: 115EH**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year I Semester Examinations, February - 2022**
**OPERATING SYSTEMS**
**(Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Answer any FIVE Questions**
**All Questions carry Equal Marks**
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1.a) What are the main differences between operating systems for mainframe computers and personal computers?
b) With a neat sketch explain the structure of OS. [8+7]

2.a) Define operating system goals from user's view and system's view.
b) What are the different types of operating systems? Discuss in detail. [7+8]

3.a) Explain the concept of semaphores. Illustrate with an example.
b) Give Peterson solution for critical section problem. [8+7]

4.a) Discuss in detail about Thread scheduling.
b) Explain how producer-consumer problem is solved using semaphores with example pseudo code. [7+8]

5.a) A process refers to 5 pages, A, B, C, D, and E in the order- A; B; C; D; A; B; E; A; B; C; D; E. If the page replacement algorithm is LRU, calculate the number of page faults with empty frames of size 4?
b) Explain in detail about paging technique. [8+7]

6. Consider the following page reference strings:
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6.
How many page faults would occur for the following replacement algorithm, assuming three, four frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
a) LRU replacement
b) Optimal replacement. [15]

7. Explain in detail about different file allocation methods. [15]

8. A system has 3 devices D1, D2 and D3 and 3 processes P1, P2, and P3. P1 is holding D1 and waiting for D3. P2 is holding D2 and waiting for D1. P3 is holding D3 and waiting for D2. Draw resource allocation graph and wait-for graph. Is the system in deadlock state or not? Explain. [15]

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user paper feb-22 intu